Frequency-Aware Kernel-Centered Prior Attention Network for Hyperspectral Image Classification

Xiangyu Wang, Shuli Cheng, Anyu Du, and Yao Ding

***Abstract*—Hyperspectral image (HSI) classification remains challenging because local discriminative cues around land-cover transitions, mixed pixels, and abrupt material boundaries are often under-modeled, while global attention can be distracted by spatially disconnected yet spectrally similar regions. To address these issues, this article proposes a frequency-aware kernel-centered prior attention network (FKCPA-Net) for HSI classification. The network first builds multiscale spectral–spatial representations to capture spectral continuity at different granularities. It then enhances local discrimination through a frequency-aware spectral–spatial fusion strategy that combines 3-D discrete cosine transform (DCT) high-pass enhancement, channel recalibration, and cross-group spectral–spatial interaction, thereby strengthening boundary, texture, and mixed-pixel representations. To improve global context modeling, a kernel-centered prior attention (KCPA) mechanism introduces complementary structural priors and a relative spatial-distance bias into dual-path attention and selectively preserves strongly correlated token interactions during normalization. Experiments on the Indian Pines (IPs), Botswana (BS), WHU-Hi-LongKou (LK), and HuangHeKou (HHK) datasets yield overall accuracies of 99.195%, 99.846%, 98.992%, and 98.111%, respectively. Additional analyses show that FKCPA-Net produces more spatially coherent classification maps and remains robust under limited training samples, demonstrating its effectiveness for complex HSI scenes. The code is available at https://github.com/DEVM369/FKCPA-Net**

***Index Terms*—Attention mechanism, frequency-aware modeling, hyperspectral image (HSI) classification, spectral–spatial feature learning, structural prior.**

I. Introduction

HYPERSPECTRAL images (HSIs) measure scene reflectance over dozens to hundreds of contiguous narrow bands, providing coupled spectral signatures and spatial context for fine-grained remote sensing analysis [1]. Compared with conventional multispectral imagery, HSIs are more effective in distinguishing materials with subtle spectral differences and have therefore been widely used in land-cover mapping, precision agriculture, mineral exploration, and environmental monitoring [2], [3]. However, accurate HSI classification is still challenged by severe spectral redundancy, strong interband correlation, limited labeled samples, mixed pixels, and irregular spatial distributions of ground objects [4]. These characteristics make robust spectral–spatial representation learning a central issue in HSI interpretation.

Early studies on HSI classification mainly relied on traditional machine learning methods, including support vector machines (SVMs) [5], random forests (RFs) [6], and manifold-preserving or dimensionality-reduction-based classification strategies [7]. These representative methods generally perform classification based solely on pixel-level spectral vectors. However, their performance is highly dependent on handcrafted feature design, and the primary emphasis on 1-D spectral information limits the exploitation of spatial contextual relationships, thereby constraining classification accuracy in complex scenes [8]. Although dimensionality reduction techniques such as principal component analysis (PCA) [9] and linear discriminant analysis (LDA) [10] were subsequently introduced to mitigate the curse of dimensionality, the inherent limitations of manual feature engineering still hinder the full utilization of deep spectral–spatial coupling information in HSI data.

With the advancement of deep learning, convolutional neural networks (CNNs) have been widely introduced into HSI classification owing to their local connectivity and weight-sharing properties. Early studies employed 1-D CNNs [11] to automatically learn features from pixelwise spectral vectors. Subsequently, 2-D CNNs [12] were investigated for spatial feature extraction, and 3-D-CNNs [13] were further developed for joint spectral–spatial representation learning. In particular, HybridSN, proposed by Roy et al. [14], captures spectral–spatial features through a cascade of 3-D and 2-D convolutional layers, thereby improving classification performance while reducing the number of parameters. In addition, considerable attention has been devoted to integrating multiscale convolutional structures and attention mechanisms with CNNs. For example, Liu et al. [15] proposed MSLKCNN, a multiscale large-kernel CNN that captures spectral–spatial features at local and global scales. Haut et al. [16] proposed a dual-branch attention module in which discriminative spectral–spatial features are extracted through multiplicative interactions between attention masks and convolutional features. The difficulty of modeling long-range dependencies with local convolutional feature extractors has motivated

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hybrid designs such as SSFTT [17]. Consequently, CNN-based methods tend to encounter performance limitations when large-scale or spatially complex ground objects are involved.

The introduction of transformers has established an alternative paradigm for HSI classification. Because multihead self-attention can model long-range dependencies directly, transformer-based architectures have been increasingly adopted for hyperspectral spectral–spatial feature learning [18]. Hong et al. [19] proposed SpectralFormer, which treats HSI data along the spectral dimension as sequential data and exploits grouped spectral embedding together with cross-layer skip connections to capture fine-grained local spectral representations within a transformer encoder, thereby facilitating the broader application of transformers in HSI classification. Jiang et al. [20] proposed CPFormer, which employs center-pixel guided dual-path attention to jointly model spatial–spectral dependencies and perform nonlinear feature fusion. However, standard self-attention considers all query-key pairs and relies mainly on feature similarity, without explicitly encoding spatial geometric priors [21]. In HSI classification, weakly related or irrelevant tokens may still contribute to attention normalization, which can dilute the weights assigned to informative regions and weaken the stability of global context modeling.

To jointly exploit the strengths of CNNs in local feature extraction and transformers in global dependency modeling, CNN-transformer hybrid architectures have gradually emerged as an important direction in HSI classification. Cheng et al. [22] proposed a hybrid covariance attention and cross-layer fusion transformer network (CACFTNet), which adopts a dual-branch architecture to capture local spectral–spatial features and global dependency information using CNNs and transformers, respectively. Yang et al. [23] proposed ACTN, which couples a CNN branch and a transformer branch via adaptive response fusion to jointly learn local contextual features and global dependency representations. Yang et al. [24] proposed S2CIFTNet, which leverages multiscale spatial–spectral coupling and interactive attention fusion to jointly capture local details, nonlinear interband relationships, and global context. Zhang et al. [25] proposed CentralGraphFormer, which integrates spectral continuity modeling, center-guided spatial screening, and sparse graph reasoning for efficient spectral–spatial classification. Recent studies have further emphasized scale-related and local-to-global representation learning in hyperspectral interpretation. Chen et al. [26] proposed SiT, which introduces scale interaction into transformer-based HSI classification. Chen et al. [27] developed MDA-HTD to explore mask-driven spectral learning for hyperspectral target detection, and Chen et al. [28] also proposed TL2GH^{2}T for local-to-global hyperspectral change detection. These studies further demonstrate the importance of multigranularity contextual modeling. Although CNN-transformer hybrid architectures enhance local–global complementarity, simple concatenation or shallow fusion usually provides only coarse feature aggregation. This may weaken high-frequency cues around boundaries, textures, and mixed pixels, while similarity-based attention can still be affected by spatially disconnected but spectrally similar regions. These limitations motivate frequency-aware kernel-centered prior attention network (FKCPA-Net), which couples frequency-aware spectral–spatial enhancement with kernel-centered prior attention (KCPA) for integrated local–global representation learning.

Frequency-domain transformation provides an explicit mechanism for decomposing features into different frequency components and therefore offers a natural means of modeling the aforementioned high-frequency discriminative cues. From the perspective of ground-scene correspondence, low-frequency components mainly describe smooth and homogeneous responses within land-cover regions, whereas high-frequency components are more associated with abrupt spectral–spatial variations, such as land-cover transitions, material boundaries, texture details, and mixed pixels [29]. These regions usually contain subtle local differences and spectral ambiguity, making them difficult to distinguish in fine-grained classification. Therefore, enhancing high-frequency spectral–spatial cues can strengthen local discriminative representations and improve the model's sensitivity to boundary and mixed-pixel regions. Consequently, frequency-domain modeling has gradually become a complementary perspective in hyperspectral feature learning. Fan et al. [30] proposed a frequency-topology interactive network that extracts both high- and low-frequency features in the frequency domain to enhance class discrimination. He et al. [31] designed a frequency-domain information enhancement module to extract structural shapes and texture details under semantic constraints in the frequency phase. Qiao and Huang [32] proposed a hierarchical dual-frequency transformer network that captures high- and low-frequency features through a dual-branch architecture. In addition, DCTransformer [33] combines channel attention with the discrete cosine transform (DCT) to extract spectral–spatial features and preserve information across a wider frequency range. Cao et al. [34] proposed DTM, which integrates DCT-based frequency-domain regularization with transformer-Mamba modeling to improve spectral–spatial representation learning for HSI classification. However, these DCT-based approaches do not explicitly construct 3-D spectral–spatial high-frequency priors for boundary, texture, and mixed-pixel discrimination. MSA-LWFormer [35] employs the fast Fourier transform (FFT) within a transformer architecture to map features from the spatial domain to the frequency domain, thereby replacing the dense computation of standard self-attention and reducing time complexity. Nevertheless, its frequency-domain operations are primarily intended to improve computational efficiency rather than to enhance feature discriminability. Taken together, existing frequency-domain and attention-based HSI classification methods still have two limitations. First, DCT-based channel recalibration and regularization, Fourier-domain global filtering, and frequency-spatial fusion methods mainly use transform-domain representations for recalibration, regularization, filtering, or efficient modeling, while explicit 3-D spectral–spatial high-frequency priors for boundary, texture, and mixed-pixel discrimination remain insufficiently explored. Second, center-pixel-guided and dual-path attention methods improve spatial–spectral interaction, but they mainly rely on

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feature similarity and rarely regularize the complete token-to-token correlation structure with explicit spatial geometry.

To address these limitations, this article presents FKCPA-Net, which couples frequency-aware spectral–spatial shuffle fusion (FASSF)-based 3-D spectral–spatial high-frequency enhancement with KCPA-based spatially constrained attention. Specifically, FASSF strengthens boundary, texture, and mixed-pixel representations by constructing high-frequency spectral–spatial priors, while KCPA introduces kernel centering, relative spatial-distance bias, and selective normalization to learn more discriminative and spatially consistent global dependencies. The main contributions of this article are summarized as follows.

1) We propose FKCPA-Net, a unified HSI classification framework that combines multiscale spectral–spatial embedding, frequency-aware local enhancement, and prior-guided global attention.
2) We design a FASSF module that integrates 3-D DCT-based high-frequency enhancement, channel recalibration, and spectral–spatial channel shuffling to better capture boundaries, textures, and mixed-pixel details.
3) We develop a KCPA module that introduces kernel-centered structural priors, a relative spatial-distance bias, and Selective Softmax to suppress weakly related tokens and improve global dependency modeling.

The remainder of this article is organized as follows. Section II reviews related work. Section III details the proposed method. Section IV presents the experimental results and analysis. Finally, Section V concludes this article.

II. Related Work

A. CNN-Based HSI Classification Methods

CNNs have been widely applied to HSI classification owing to their effectiveness in local feature extraction. For 1-D CNNs, Zhai et al. [36] proposed a dual-attention-based multilevel 1-D-CNN to strengthen the discriminative representation of spectral features. However, such methods remain primarily confined to feature learning along the spectral dimension, and spatial contextual information is only partially exploited. To alleviate this limitation, 2-D CNNs were subsequently investigated for spatial feature extraction. Makantasis et al. [37] combined PCA with a 2-D CNN and extracted spatial structural features from dimension-reduced neighborhood blocks, resulting in improved classification performance. Nevertheless, because 2-D CNNs operate on individual slices, the 3-D spectral–spatial structure of HSI data cannot be fully utilized. Consequently, 3-D-CNNs have received increasing attention. Ahmad et al. [38] proposed a lightweight 3-D-CNN that extracts joint spectral–spatial features from small overlapping 3-D patches, thereby reducing computational cost while preserving classification accuracy. Zhong et al. [39] developed a spectral–spatial residual network (SSRN), in which spectral correlations and spatial contextual features are learned separately through spectral and spatial residual blocks, with identity mappings used to mitigate gradient degradation in deep networks. Multiscale feature learning has also been widely investigated. Paoletti et al. [40] proposed a pyramidal residual network to capture spectral–spatial features at multiple granularities by means of multiscale convolutional kernels, thereby improving the representation of complex land-cover classes. Gao et al. [41] designed a multiscale residual hybrid depthwise separable convolutional network to extract multigranularity spectral–spatial features through hybrid depthwise convolution and multiscale residual connections. Song et al. [42] proposed a deep feature fusion network, through which shallow detailed features and deep semantic features are effectively integrated via a densely connected architecture. Li et al. [43] proposed a model-guided coarse-to-fine fusion network that progressively reconstructs high-resolution HSIs through coarse fusion and fine refinement. Li et al. [44] developed a model-informed multistage unsupervised network that combines degradation information learning with progressive image reconstruction for HSI super-resolution. Li et al. [45] proposed an enhanced deep image prior framework that improves unsupervised HSI super-resolution through input initialization and internal prior learning. Overall, CNN-based methods exhibit clear advantages in local texture characterization and neighborhood context utilization. However, feature interactions in these methods remain largely constrained by fixed convolutional kernels and local receptive fields, thereby limiting the modeling of long-range dependencies, the maintenance of cross-region consistency for homogeneous land-cover classes, and the achievement of global discrimination in regions with complex boundaries.

B. Transformer-Based HSI Classification Methods

Transformer-based methods have been increasingly adopted in HSI classification because self-attention mechanisms facilitate the modeling of long-range spectral–spatial dependencies. Mei et al. [46] proposed a group-aware hierarchical transformer (GAHT), in which adjacent pixels are grouped and embedded to reduce sequence length, thereby improving spatial context modeling while retaining computational efficiency. Peng et al. [47] developed a spatial–spectral cross-attention transformer to strengthen feature interaction between the spectral and spatial domains through cross-attention between the two branches. Song et al. [48] designed a bottleneck spatial–spectral transformer (BS2T), in which the token dimension is compressed through a bottleneck structure to improve computational efficiency while preserving key spectral–spatial features. Zhao et al. [49] proposed HGTFN, which integrates a key-pixel guided transformer with adaptive multihop hypergraph construction to jointly capture local structural relations and long-range spectral–spatial dependencies. In addition, Jia et al. [50] proposed a center-masked transformer (CMT), which improves HSI classification under limited training samples by introducing a local-to-global token embedding framework, a multiscale convolutional token embedding module, and a regularized center-masked pretraining task, thereby enhancing local–global representation learning and the utilization of unlabeled samples. Guo et al. [51] proposed PCRNet, which introduces memory-guided global attention in the correlation space and progressively refines correlation maps for few-shot HSI classification. Despite these advances, many transformer-based and correlation-refinement

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methods still rely on dense or weakly constrained attention and do not sufficiently encode spatial geometric structure. Consequently, weakly correlated or irrelevant tokens may still participate in feature aggregation, which can blur classification decision boundaries.

C. Frequency-Domain Feature Learning Methods

Frequency-domain transformation enables the explicit decomposition of image features into components with different frequencies, thereby providing a complementary perspective for feature learning in HSI classification. Rao et al. [52] proposed the global filter network (GFNet), in which self-attention in the spatial domain is replaced by learnable global filters in the frequency domain, enabling global feature interaction with log-linear complexity and a favorable balance between efficiency and accuracy. For HSI classification, Shi et al. [53] proposed a fast Fourier filter network (F3Net), which introduces the FFT into HSI classification and efficiently mixes spectral–spatial feature information through frequency-domain fast Fourier filters. Shi et al. [54] further developed a multiscale hierarchical convolution-assisted FourierFormer (MHCFormer), which effectively integrates CNNs, transformers, and the Fourier transform for HSI classification, enabling global–local collaborative modeling through multiscale spectral–spatial token generation and hierarchical token learning. Kavalerov et al. [55] proposed a 3-D Fourier scattering transform method to extract frequency-domain features of hyperspectral data through multilayer scattering operations in the 3-D Fourier domain. Wang et al. [56] investigated the use of frequency-domain features for HSI classification and extracted frequency-domain descriptors through the discrete Fourier transform to enhance land-cover discrimination. For joint learning across the frequency and spatial domains, Pan et al. [57] proposed a frequency- and spatial-domain feature fusion network (FSFF-Net), which adopts a dual-branch parallel architecture composed of a frequency-domain transformer and deep convolution to capture frequency-domain global features and spatial-domain local features, respectively. Collectively, these studies indicate that frequency-domain modeling can provide complementary discriminative cues related to edges, textures, and structural variations. Nevertheless, compared with DCTransformer [33], which mainly combines DCT with channel attention for transform-domain feature modeling, FASSF further performs 3-D DCT-based spectral–spatial high-frequency enhancement, integrates it with channel shuffling and local 3-D fusion, and uses dual-descriptor reweighting attention (DDRA) for adaptive channel recalibration. Therefore, FASSF focuses more on joint spectral–spatial high-frequency prior modeling and fine-grained boundary, texture, and mixed-pixel representation.

III. Methodology

This section presents the architecture of FKCPA-Net in detail. To avoid notation ambiguity, let L denote the number of spectral bands in an input hyperspectral patch, let C denote the number of channels in the intermediate network features, and let $H \times W$ denote the spatial size. For intermediate features, H', W', and L' denote the spatial height, spatial width, and spectral dimension of intermediate features, respectively. Given an input patch $\mathbf{X} \in \mathbb{R}^{H\times W\times L}$, the network successively performs shallow spectral–spatial projection, the multiscale spectral feature extraction (MSPeFE) module, the FASSF module, and the KCPA module, followed by a classification head that outputs class probabilities.

A. Overview of FKCPA-Net

As illustrated in Fig. 1, FKCPA-Net adopts a hierarchical design that proceeds from shallow to deep layers and from local to global modeling.

First, the input patch is processed by a shallow spectral–spatial projection stem, which consists of a Conv3D layer with C output channels, a kernel size of $1 \times 1 \times 7$, a stride of $1 \times 1 \times 2$, and no padding, followed by an AvgPool3D layer with a kernel size of $1 \times 1 \times 2$ and a stride of $1 \times 1 \times 2$, yielding the shallow feature $\mathbf{F}_0$. Next, $\mathbf{F}_0$ is fed into the MSPeFE module, which uses multiscale spectral convolutions to capture local spectral continuity at different granularities and produces the multiscale embedded feature $\mathbf{F}_{ms}$. Based on this representation, the FASSF module enhances local discriminative cues from two complementary perspectives: one branch explicitly extracts spectral–spatial high-frequency priors in the frequency domain, whereas the other promotes intergroup information exchange via spectral–spatial channel shuffling. Their fusion yields the enhanced feature $\mathbf{F}_{sf}$. The KCPA module then compresses $\mathbf{F}_{sf}$, reshapes it into a sequence of spatial tokens, and performs global modeling using dual-path complementary attention, cross-path kernel-centered prior modulation, and a spatial-distance bias, followed by final classification. Unlike parallel addition or concatenation, the FASSF–KCPA cascade follows a content-to-relation order: FASSF first enhances local high-frequency spectral–spatial cues, and KCPA then constructs spatial tokens and attention relations from the enhanced feature. Therefore, the frequency-aware local evidence directly conditions the subsequent geometrically constrained global modeling, rather than being fused only after independent branch processing. For reproducibility, the implementation uses $C = 32$ feature channels throughout the main feature-extraction and attention modules. FKCPA-Net contains one shallow spectral–spatial projection stem, one MSPeFE block, one FASSF block, one KCPA block, and one final classification head.

B. MSPeFE Module

The MSPeFE block proposed in [58] is adopted as the multiscale spectral embedding stem of FKCPA-Net. For completeness, its operation is briefly summarized as follows. The shallow feature $\mathbf{F}_0 \in \mathbb{R}^{C\times H' \times W' \times L'}$ is evenly partitioned into four groups along the channel dimension. Each group is then processed using a 3-D depthwise separable convolution with a spectral kernel size of $1 \times 1 \times k_i$, where $k_i \in \{3, 7, 11, 15\}$. The resulting feature maps are concatenated to obtain the multiscale embedded feature $\mathbf{F}_{ms}$, which is subsequently used as the input to the proposed FASSF and KCPA modules.

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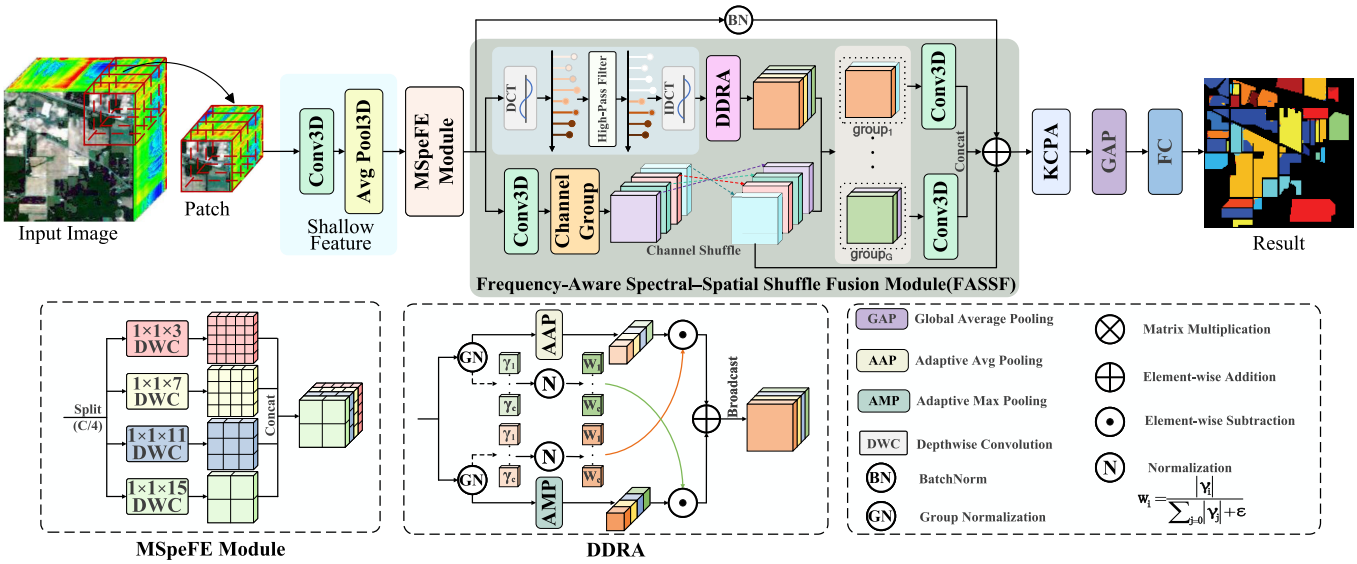


Fig. 1. Proposed FKCPA-Net network architecture. The network consists of shallow spectral–spatial projection, the MSPeFE module, the FASSF module, and the KCPA module.

C. FASSF Module

Although $\mathbf{F}_{ms}$ already contains multiscale spectral information, it still lacks explicit modeling of high-frequency cues related to land-cover transition zones, mixed pixels, and abrupt material changes. Meanwhile, relying solely on grouped operations may restrict intergroup information interaction. To address these issues, we design a FASSF module that improves local discriminative representation through coordinated frequency enhancement and spectral–spatial channel shuffling.

Given the input feature $\mathbf{F}_{ms}$, the frequency-enhancement branch first performs a separable 3-D DCT to obtain the frequency-domain representation

$$\begin{aligned}\mathbf{Z} &= \mathrm{DCT}_{3-D}\left(\mathbf{F}_{ms}\right) \\ &= \mathrm{DCT}^{(L')}\left(\mathrm{DCT}^{(W')}\left(\mathrm{DCT}^{(H')}\left(\mathbf{F}_{ms}\right)\right)\right)\end{aligned} \tag{1}$$

where $\mathrm{DCT}^{(H')}$, $\mathrm{DCT}^{(W')}$, and $\mathrm{DCT}^{(L')}$ denote 1-D DCTs along the spatial height, spatial width, and spectral dimensions, respectively. Compared with the FFT, the DCT yields only real-valued coefficients and exhibits better energy compaction, making it more suitable for separating low- and high-frequency information by means of a simple mask.

To emphasize high-frequency responses, a high-pass mask $\mathbf{M}_{hp}$ is constructed to suppress the low-frequency corner region of the spectrum

$$\widehat{\mathbf{Z}} = \mathbf{Z} \odot \mathbf{M}_{hp} \tag{2}$$

where $\odot$ denotes elementwise multiplication. Let τ denote the cutoff ratio. The mask is defined as

$$\mathbf{M}_{hp}(i,j,l) = \begin{cases} 0, & i < \lfloor \tau H' \rfloor,\ j < \lfloor \tau W' \rfloor,\ l < \lfloor \tau L' \rfloor \\ 1, & \text{otherwise.} \end{cases} \tag{3}$$

The filtered coefficients are then transformed back to the original domain to obtain the high-frequency prior feature

$$\mathbf{F}_h = \mathrm{IDCT}_{3-D}\left(\widehat{\mathbf{Z}}\right). \tag{4}$$

DCT_{3-D} and IDCT_{3-D} are applied independently to each channel over the H', W', and L' dimensions; no transform is performed across channels. The reconstructed high-frequency feature $\mathbf{F}_h$ may contain both discriminative details and noise. We therefore introduce DDRA to adaptively recalibrate its channels. Specifically, two independent group-normalization branches generate complementary views of $\mathbf{F}_h$. Adaptive average pooling and adaptive max pooling are then applied to the two views to obtain the channel descriptors

$$\mathbf{d}_a = \mathrm{AAP}\left(\mathrm{GN}_a\left(\mathbf{F}_h\right)\right), \quad \mathbf{d}_m = \mathrm{AMP}\left(\mathrm{GN}_m\left(\mathbf{F}_h\right)\right). \tag{5}$$

Meanwhile, the learnable scaling parameters of the two group-normalization branches, denoted by $\boldsymbol{\gamma}_a$ and $\boldsymbol{\gamma}_m$, are used to compute normalized channel-wise weights

$$\mathbf{w}_a = \frac{|\boldsymbol{\gamma}_a|}{\sum_c |\gamma_a^c| + \epsilon}, \quad \mathbf{w}_m = \frac{|\boldsymbol{\gamma}_m|}{\sum_c |\gamma_m^c| + \epsilon} \tag{6}$$

where ϵ is a small constant used to prevent division by zero.

Adaptive average pooling summarizes the overall response of each channel, whereas adaptive max pooling emphasizes its strongest activation. We therefore cross-weight the two descriptors so that each descriptor calibrates the other. This operation reduces the risk of repeatedly amplifying redundant channel responses

$$\mathbf{r} = \sigma\left(\mathbf{d}_a \odot \mathbf{w}_m + \mathbf{d}_m \odot \mathbf{w}_a\right) \tag{7}$$

where $\sigma(\cdot)$ denotes the sigmoid function. The output of the frequency-enhancement branch is thus given by

$$\mathbf{X}_f = \mathbf{r} \odot \mathbf{F}_h \tag{8}$$

where $\mathbf{r}$ is broadcast along the spatial and spectral dimensions to match the size of $\mathbf{F}_h$.

In parallel, the spectral–spatial branch first applies a $1 \times 1 \times 1$ 3-D convolution to linearly project the input feature in the channel space

$$\widetilde{\mathbf{X}}_p = \mathrm{Conv3D}\left(\mathbf{F}_{ms}\right). \tag{9}$$

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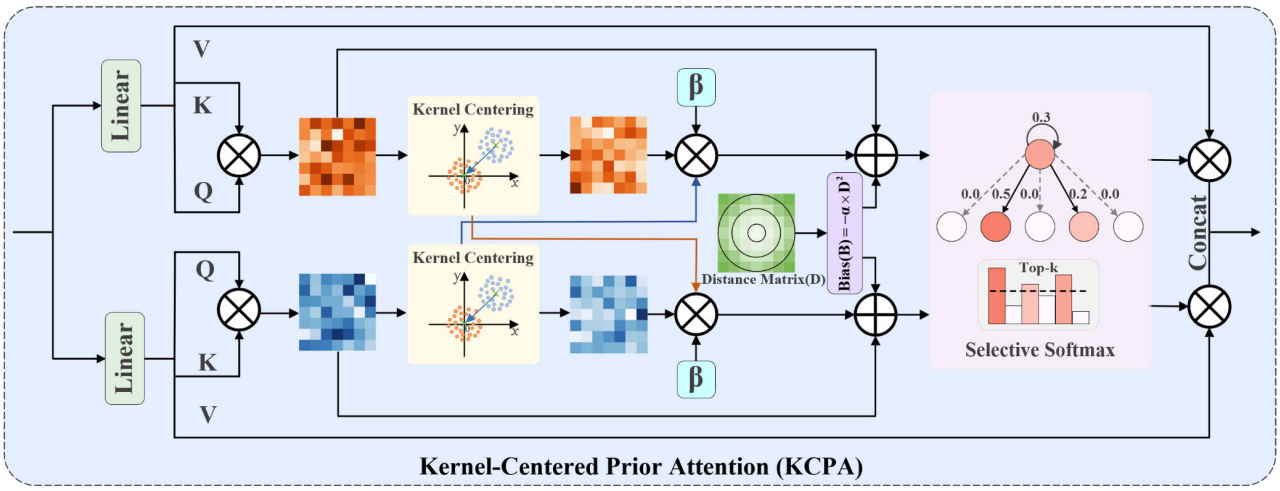


Fig. 2. Schematic of the KCPA module.

In the implementation, the channel projection in (9) is performed using a $1 \times 1 \times 1$ Conv3D layer followed by BatchNorm3D and ReLU6 activation. Next, $\widetilde{\mathbf{X}}_p$ is evenly divided into G groups along the channel dimension, where G denotes the number of channel groups. Channel shuffling is then performed to alleviate the information isolation caused by grouping and to promote cross-group information flow, yielding the shuffled feature $\mathbf{X}_p$. During fusion, $\mathbf{X}_f$ and $\mathbf{X}_p$ are split into G corresponding groups. For each group, the frequency-enhanced and shuffled features are concatenated and then passed through an intragroup 3-D convolution for local spectral–spatial fusion

$$\mathbf{U}^{(g)} = \text{Conv3D}\left(\text{Concat}\left(\mathbf{X}_f^{(g)}, \mathbf{X}_p^{(g)}\right)\right), \quad g = 1, \ldots, G. \tag{10}$$

The convolutional fusion in (10) is implemented using $1 \times 1 \times 1$ Conv3D layers followed by BatchNorm3D and ReLU6 activation. Finally, the group outputs are concatenated and combined with BN($\mathbf{F}_{ms}$) and $\mathbf{X}_p$ through residual addition to obtain the final output of FASSF

$$\mathbf{F}_{sf} = \text{BN}\left(\mathbf{F}_{ms}\right) + \text{Concat}\left[\mathbf{U}^{(1)}, \mathbf{U}^{(2)}, \ldots, \mathbf{U}^{(G)}\right] + \mathbf{X}_p. \tag{11}$$

Through this design, FASSF simultaneously performs frequency-domain high-frequency detail enhancement, cross-group information flow, and local spectral–spatial interaction within a unified module, which improves the network's sensitivity to boundaries, textures, and local anomalous variations.

D. Kernel-Centered Prior Attention

FASSF enhances local fine-grained features but does not explicitly model long-range spatial dependencies. We therefore apply KCPA sequentially to F_{sf}, as illustrated in Fig. 2, so that its spatial tokens and attention correlations are constructed from the frequency-enhanced representation. After spectral compression, KCPA applies global attention to these tokens and regularizes the attention logits using cross-path kernel-centered priors and a relative spatial-distance bias. Thus, FASSF strengthens local discriminative evidence, while KCPA constrains its global propagation according to structural correlation and spatial geometry. The following formulation describes these operations in a multihead attention setting.

Given the output $\mathbf{F}_{sf}$ of FASSF, a $1 \times 1 \times L'$ Conv3D layer is first applied along the spectral dimension to compress it into $\mathbf{F}_{in} \in \mathbb{R}^{C \times H' \times W' \times 1}$. The feature is then flattened along the spatial dimensions into a token sequence $\mathbf{T} \in \mathbb{R}^{N \times C}$, where $N = H' \times W'$, and each token corresponds to a C-dimensional fused feature at one spatial location. Two independent linear mappings are then used to construct dual-path representations

$$\mathbf{M}_1 = \mathbf{T}\mathbf{W}_{m_1}, \qquad \mathbf{M}_2 = \mathbf{T}\mathbf{W}_{m_2} \tag{12}$$

where $\mathbf{W}_{m_1}$ and $\mathbf{W}_{m_2}$ are learnable parameter matrices. For each path, multihead query, key, and value matrices are computed. For the first path

$$\mathbf{Q}_1 = \mathbf{M}_1\mathbf{W}_{Q_1}, \qquad \mathbf{K}_1 = \mathbf{M}_1\mathbf{W}_{K_1}, \qquad \mathbf{V}_1 = \mathbf{M}_1\mathbf{W}_{V_1}. \tag{13}$$

The query, key, and value projections preserve the embedding dimension C. These matrices are reshaped into multihead form, i.e., $\mathbf{Q}_1, \mathbf{K}_1, \mathbf{V}_1 \in \mathbb{R}^{h \times N \times d}$, where h is the number of attention heads and $d = C/h$ is the dimension of each head. In our implementation, $C = 32$ and $h = 2$; thus, each attention head has $d = 16$ dimensions. The second path is computed in the same manner. The attention correlation matrices of the two paths are then given by

$$\mathbf{R}_1 = \frac{\mathbf{Q}_1\mathbf{K}_1^\top}{\sqrt{d}}, \quad \mathbf{R}_2 = \frac{\mathbf{Q}_2\mathbf{K}_2^\top}{\sqrt{d}}. \tag{14}$$

To enhance the complementarity between the two paths, we introduce a cross-path kernel-centered prior modulation mechanism inspired by kernel matrix centering. Specifically, kernel centering is applied to the correlation matrix of each attention head

$$\widehat{\mathbf{R}} = \mathbf{M}_{\text{cen}}\mathbf{R}\mathbf{M}_{\text{cen}}, \quad \mathbf{M}_{\text{cen}} = \mathbf{I}_{\text{N}} - \frac{1}{N}\mathbf{1}\mathbf{1}^\top \tag{15}$$

where $\mathbf{R}$ denotes the input attention logit matrix, $\mathbf{M}_{\text{cen}}$ is the kernel-centering matrix, $\mathbf{I}_{\text{N}}$ is the $N \times N$ identity matrix, and $\mathbf{1}$ is the all-ones vector. Different from standard layer normalization, which normalizes channel-wise feature activations,

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kernel centering is directly performed on the token-to-token attention correlation matrix. This operation removes the shared mean-correlation component among tokens and highlights relative correlation structures. In HSI scenes, spatially disconnected regions may still show similar spectral responses and produce globally high attention scores. By suppressing this common spectral-similarity bias, the kernel-centered prior encourages attention to focus on spatially coherent and structurally consistent token relations.

On this basis, the kernel-centered result from the other path is injected as a complementary prior into the current path to modulate its attention logit matrix

$$\widetilde{\mathbf{R}}_1 = \mathbf{R}_1 + \beta \odot \widehat{\mathbf{R}}_2 \quad \widetilde{\mathbf{R}}_2 = \mathbf{R}_2 + \beta \odot \widehat{\mathbf{R}}_1 \tag{16}$$

where β is a learnable head-wise scaling parameter. This cross-path kernel-centered prior modulation preserves the original correlation measure of each path while introducing the kernel-centered structural information from the other path as a complementary prior, thereby improving the discriminability of the attention distribution for complex land-cover categories.

Cross-path kernel-centered prior modulation alone is still insufficient to guarantee consistency between attention and spatial geometry. Therefore, we further introduce a distance prior based on 2-D spatial positions. Let $\mathbf{s}_i = (x_i, y_i)$ and $\mathbf{s}_j = (x_j, y_j)$ denote the spatial coordinates of the ith and jth tokens, respectively. The Euclidean distance matrix is defined as

$$D_{ij} = \left\| \mathbf{s}_i - \mathbf{s}_j \right\|_2. \tag{17}$$

Because land-cover classes in HSI usually form connected regions and nearby pixels are more likely to share material composition, the 2-D spatial distance provides a physically meaningful measure of token proximity. Based on this observation, a learnable additive bias is constructed as

$$B_{dis} = -\alpha D_{ij}^2 \tag{18}$$

where α is a nonnegative learnable parameter, and D^2 denotes elementwise squaring. This quadratic decay makes the bias decrease monotonically with increasing distance and explicitly increases the attention weight of spatially neighboring tokens at the logit level, thus encouraging local spatial consistency in the attention distribution. Compared with absolute positional encoding, the relative-distance-based bias is naturally translation-invariant and can better adapt to objects appearing at different positions and scales. This bias term is added to the attention logit matrices after cross-path kernel-centered prior modulation

$$\mathbf{R}_1' = \widetilde{\mathbf{R}}_1 + \mathbf{B}_{dis}, \quad \mathbf{R}_2' = \widetilde{\mathbf{R}}_2 + \mathbf{B}_{dis}. \tag{19}$$

During normalization, a Selective Softmax is adopted to suppress the interference of weakly correlated tokens on attention allocation. For each query position, only the top-k most relevant key-value pairs are retained

$$\begin{aligned} \mathbf{A}_1 &= \mathrm{Softmax}\left(\mathrm{mask}\left(\mathbf{R}_1', k\right)\right) \\ \mathbf{A}_2 &= \mathrm{Softmax}\left(\mathrm{mask}\left(\mathbf{R}_2', k\right)\right) \end{aligned} \tag{20}$$

where $k = \lfloor \rho N \rfloor$, ρ is the selection ratio, and $\mathrm{mask}(\cdot, k)$ sets the values of nontop-k positions to $-\infty$. In addition, to preserve self-loop information in self-attention, the diagonal positions are always retained. This design is intended to improve attention selectivity rather than reduce the asymptotic complexity of self-attention, because the full attention logit matrix is computed before top-k masking. Therefore, k mainly controls the number of effective nonzero token interactions after masking. The mask assigns zero attention to low-correlation token pairs. Consequently, spatially disconnected tokens contribute less to feature aggregation, while nearby and structurally consistent tokens retain greater influence. This improves the discrimination of mixed pixels and boundary regions. Overall, the three key components in KCPA play complementary roles: the kernel-centered prior highlights relative structural correlations, the spatial-distance bias enhances local spatial coherence, and Selective Softmax suppresses weakly related token interactions. These designs jointly improve the structural discrimination, spatial consistency, and selectivity of global dependency modeling.

Finally, the value vectors are aggregated according to the selective attention weights to produce the outputs of the two paths

$$\mathbf{O}_1 = \mathbf{A}_1 \mathbf{V}_1, \quad \mathbf{O}_2 = \mathbf{A}_2 \mathbf{V}_2. \tag{21}$$

The outputs of the two paths are concatenated along the channel dimension to obtain the fused feature

$$\mathbf{Y} = \mathrm{Concat}\left(\mathbf{O}_1, \mathbf{O}_2\right). \tag{22}$$

The concatenated dual-path output is projected back to 32 dimensions, followed by ReLU6 activation, dropout with a rate of 0.1, and LayerNorm. The fused feature is then mapped to the class space via global average pooling and a fully connected layer to obtain the final classification result.

IV. Experiments and Analysis

We evaluate FKCPA-Net on four widely used HSI benchmarks and compare it with representative state-of-the-art methods. This section first introduces the datasets and implementation details, and then reports parameter sensitivity, quantitative and qualitative comparisons, robustness under reduced supervision, ablation results, and attention visualization.

A. Data Description

1) Indian Pines: The Indian Pines (IPs) dataset was collected by NASA's AVIRIS sensor over northwestern Indiana, USA, in 1992. It has a spatial size of 145 × 145 pixels and covers 16 land-cover categories. The data span 0.4–2.5 μm with 220 contiguous spectral bands. After removing 20 noisy bands, 200 bands are retained for classification. Fig. 3 shows the false-color composite and the corresponding ground-truth map. The dataset contains 10 249 labeled samples, which are split into training, validation, and test subsets at a ratio of 10%, 10%, and 80%, respectively. Class-wise sample statistics are summarized in Table I.

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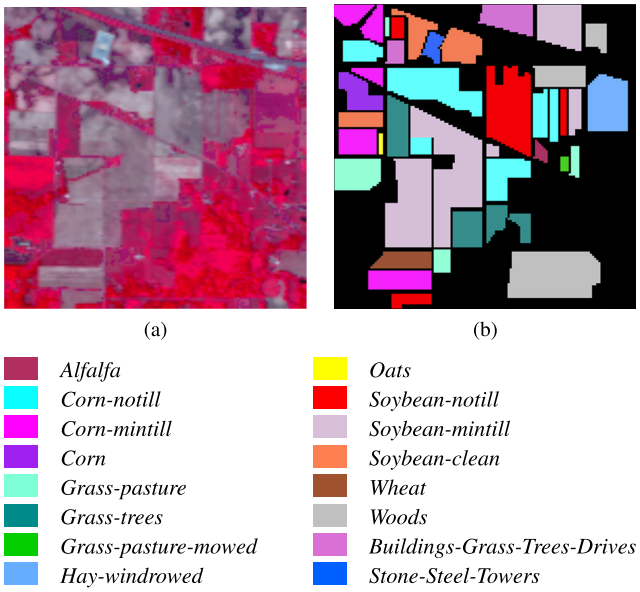


Fig. 3. IPs dataset. (a) False-color map. (b) Ground-truth map.

TABLE I
LAND COVER CATEGORIES OF THE IPS DATASET AND THE NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES

Class No.	Class name	Train	Val	Test
1	Alfalfa	5	4	37
2	Corn-notill	142	143	1143
3	Corn-mintill	83	83	664
4	Corn	24	23	190
5	Grass-pasture	48	49	386
6	Grass-trees	73	73	584
7	Grass-pasture-mowed	3	3	22
8	Hay-windrowed	48	48	382
9	Oats	2	2	16
10	Soybean-notill	97	97	778
11	Soybean-mintill	245	246	1964
12	Soybean-clean	59	59	475
13	Wheat	21	20	164
14	Woods	126	127	1012
15	Buildings-Grass-Trees-Drives	38	39	309
16	Stone-Steel-Towers	10	9	74
	Total	1024	1025	8200

2) Botswana: The Botswana (BS) dataset was acquired by NASA's EO-1 satellite over the Okavango Delta in BS in May 2001. It has a spatial size of 1476 × 256 pixels with an approximate spatial resolution of 20 m. The data cover the 0.4–2.5 μm spectral range and contain 242 bands. After removing five noisy band intervals ($1-9, 56-81, 98-101, 120-133$, and $165-186$), 145 bands are retained for classification. The scene is annotated with 14 land-cover categories, including seasonal wetlands, intermittent wetlands, and dry woodland. Fig. 4 presents the false-color composite and the corresponding ground-truth map. The dataset contains 3248 labeled samples, which are split into training, validation, and test subsets at a ratio of 10%, 10%, and 80%, respectively. Class-wise sample statistics are summarized in Table II.

3) WHU-Hi-LongKou: The WHU-Hi-LongKou (LK) dataset was collected over a town in Hubei Province, China, using an uncrewed aerial vehicle platform equipped with a hyperspectral imaging sensor. It has a spatial size of

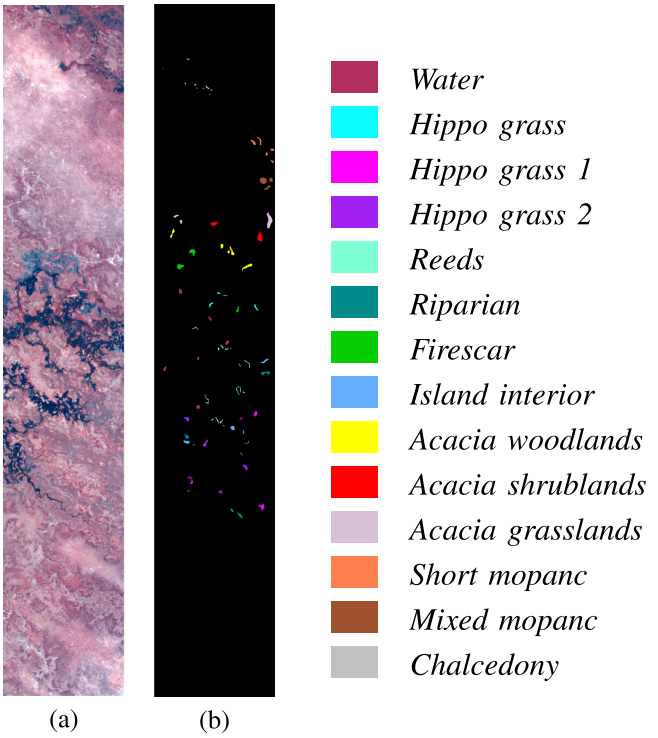


Fig. 4. BS dataset. (a) False-color map. (b) Ground-truth map.

TABLE II
LAND COVER CATEGORIES OF THE BS DATASET AND THE NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES

Class No.	Class name	Train	Val	Test
1	Water	27	27	216
2	Hippo-grass	10	10	81
3	Hippo grass-1	25	25	201
4	Hippo grass-2	22	21	172
5	Reeds	27	27	215
6	Riparian	27	27	215
7	Firescar	26	26	207
8	Island interior	21	20	162
9	Acacia woodlands	31	32	251
10	Acacia shrublands	24	25	199
11	Acacia grasslands	30	31	244
12	Short mopanc	18	18	145
13	Mixed mopanc	26	27	215
14	Chalcedony	10	9	76
	Total	324	325	2599

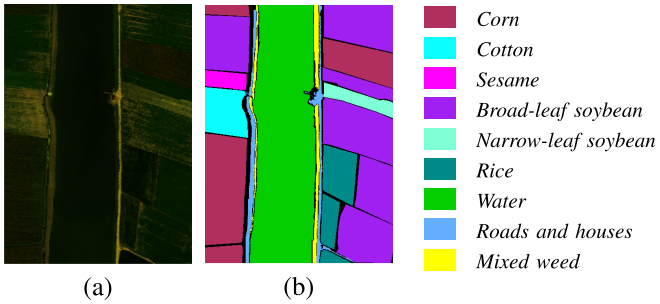


Fig. 5. WHU-Hi-LK dataset. (a) False-color map. (b) Ground-truth map.

550 × 400 pixels and covers nine land-cover categories. The data span 0.4–1.0 μm with 270 spectral bands, and the spatial resolution is approximately 0.463 m. Fig. 5 presents

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TABLE III

LAND COVER CATEGORIES OF THE WHU-HI-LK DATASET AND THE NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES

Class No.	Class name	Train	Val	Test
1	Corn	34	35	34442
2	Cotton	8	9	8357
3	Sesame	3	3	3025
4	Broad-leaf soybean	63	63	63086
5	Narrow-leaf soybean	4	4	4143
6	Rice	12	12	11830
7	Water	67	67	66922
8	Roads and houses	7	7	7110
9	Mixed weed	6	5	5218
	Total	204	205	204133

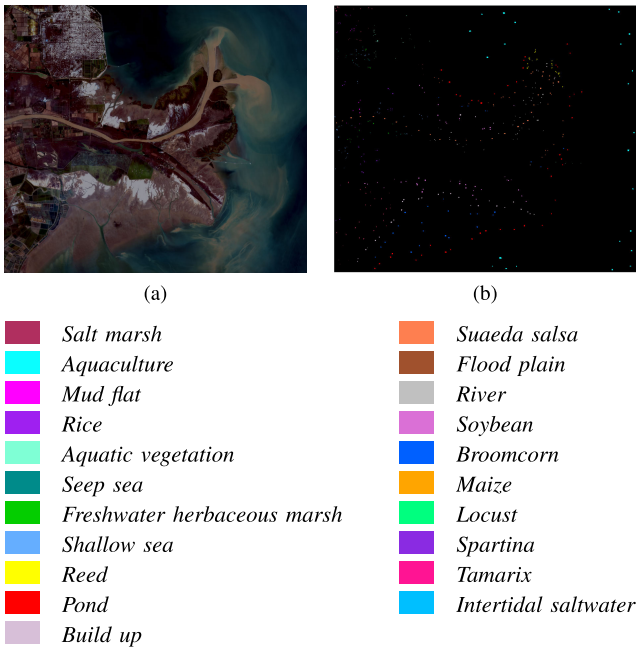


Fig. 6. HHK dataset. (a) False-color map. (b) Ground-truth map.

the false-color composite and the corresponding ground-truth map. Among the annotated categories, six correspond to major crop types—corn, cotton, sesame, broad-leaf soybean, narrow-leaf soybean, and rice—while the remaining three are water, roads and houses, and mixed weed areas. The dataset contains 204 542 labeled pixels, which are split into training, validation, and test subsets at a ratio of 0.1%, 0.1%, and 99.8%, respectively. Class-wise sample statistics are summarized in Table III.

4) HuangHeKou: The HuangHeKou (HHK) dataset was collected by GF5_AHSI over the HHK region of China in 2019. It has a spatial size of 1185 × 1342 pixels and covers 21 land-cover categories. The data span the VNIR (0.390 – 1.029 μm) and SWIR (1.005–2.513 μm) ranges, yielding 330 spectral bands in total. After removing 45 invalid bands, 285 bands are retained for classification. Fig. 6 presents the false-color composite and the corresponding ground-truth map. The dataset contains 6471 labeled samples, which are split into training, validation, and test subsets at a ratio of 5%, 5%, and 90%, respectively. Class-wise sample statistics are summarized in Table IV.

TABLE IV

LAND COVER CATEGORIES OF THE HHK DATASET AND THE NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES

Class No.	Class name	Train	Val	Test
1	Salt marsh	19	20	354
2	Aquaculture	40	40	716
3	Mud flat	6	5	99
4	Rice	10	9	171
5	Aquatic vegetation	4	4	75
6	Seep sea	5	5	86
7	Freshwater herbaceous marsh	5	5	85
8	Shallow sea	10	11	190
9	Reed	10	10	180
10	Pond	47	47	842
11	Build up	27	28	498
12	Suaeda salsa	23	24	422
13	Flood plain	18	18	325
14	River	12	12	216
15	Soybean	29	30	536
16	Broomcorn	22	23	409
17	Maize	7	6	120
18	Locust	19	19	339
19	Spartina	4	3	61
20	Tamarix	4	3	65
21	Intertidal saltwater	2	2	35
	Total	323	324	5824

B. Experimental Setup

1) Configuration: The proposed FKCPA-Net was implemented using the PyTorch framework in an experimental environment configured with Python 3.8.0 and PyTorch 2.4.1. All experiments were performed on a server equipped with an NVIDIA GeForce RTX 3090 GPU. For the general hyperparameter configuration, the Adam optimizer was employed for parameter optimization, with the weight decay coefficient set to 1×10^{-3}. The learning rate was scheduled using a cosine annealing strategy in conjunction with a warm-up mechanism. The spatial patch size of the input images was uniformly set to 9×9, and the batch size was fixed at 128. For the architectural hyperparameters, the cutoff ratio τ in the DCT-based high-frequency branch of FASSF and the selection ratio ρ in the Selective Softmax mechanism were set to 0.5 and 0.65, respectively. In addition, the number of channel groups G in FASSF was set to 4, while the number of attention heads h in KCPA was set to 2. To mitigate the effect of class imbalance in hyperspectral data, Focal Loss was adopted as the loss function. Owing to the substantial interdataset differences in land-cover distribution complexity and classification difficulty, the focusing parameter γ_{FL} and the training strategy were adjusted on a per-dataset basis. Specifically, for the IP and HHK datasets, the initial learning rate was set to 0.003, and the number of training epochs was fixed at 500, while γ_{FL} was set to 0.6 and 1.7, respectively. For the BS dataset, the learning rate was set to 0.001, the number of training epochs was fixed at 600, and γ_{FL} was set to 1.1. For the LK dataset, the learning rate was set to 0.001, the number of training epochs was fixed at 400, and γ_{FL} was set to 2.3 to place greater emphasis on hard-to-classify samples during training.

2) Evaluation Indicators: Three standard metrics were used to evaluate classification performance, namely, overall accuracy (OA), average accuracy (AA), and the Kappa coefficient

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(κ). OA measures the proportion of correctly classified test samples. AA is the mean of the class-wise accuracies and reflects the balance of performance across categories. The κ coefficient evaluates the agreement between the prediction map and the ground-truth labels after discounting chance agreement.

3) Comparison Methods: To provide a comprehensive comparison, FKCPA-Net was evaluated against representative methods from three architecture families, including CNN-based models, transformer-based models, and hybrid CNN-transformer models. The compared methods are briefly summarized below.

1) *3-D-CNN [59]:* A baseline in which raw hyperspectral cubes are processed using hierarchical 3-D convolutions and pooling for joint spectral–spatial feature extraction, followed by fully connected pixel-wise classification.
2) *S2VNet [60]:* A subpixel spectral variability network that models mixed pixels via nonlinear AE-based unmixing and endmember/abundance estimation, and improves classification through a subpixel fusion module that integrates abundance, spectral-correlation, and pixel-level class features.
3) *SQSFormer [61]:* A spectral query-spatial mechanism is introduced in a pure transformer, where the center-pixel spectrum guides neighboring-token aggregation for discriminative spatial-context modeling and linear classification.
4) *SiT [26]:* Multiscale representations are constructed in a pure transformer, and cross-scale interaction is introduced to capture local details and long-range dependencies before linear classification.
5) *CPFormer [20]:* A center-pixel-guided dual-path multihead attention fusion transformer that jointly models spatial–spectral dependencies and performs nonlinear feature fusion for HSI classification.
6) *SSFTT [17]:* Shallow spectral–spatial features are extracted by cascaded 3-D and 2-D convolutions, tokenized by a Gaussian weighted feature tokenizer, positionally encoded, and modeled by a transformer before linear classification.
7) *HybridFormer [62]:* Multigranularity spectral–spatial features are hierarchically constructed by convolution, while cross-granularity and cross-dimensional interaction are modeled in a transformer before fully connected pixel-wise classification.
8) *S2CIFTNet [24]:* Multiscale spatial–spectral coupling and interactive attention fusion are employed to jointly capture local details, nonlinear interband relationships, and global context for HSI classification.
9) *MCMTN [63]:* Mid-range convolutions are embedded into a CNN-transformer hybrid to modulate features, injecting local priors and mid-scale context into global dependency modeling before linear classification.

C. Parameter Analysis

To assess the effects of key hyperparameters on classification performance, parameter sensitivity experiments were conducted on four datasets, namely BS, IP, LK, and HHK. With all other training settings held constant, one-factor-at-a-time sensitivity analyses were performed for the patch size p, cutoff ratio τ, and selection ratio ρ. Unless otherwise stated, OA was used as the evaluation metric.

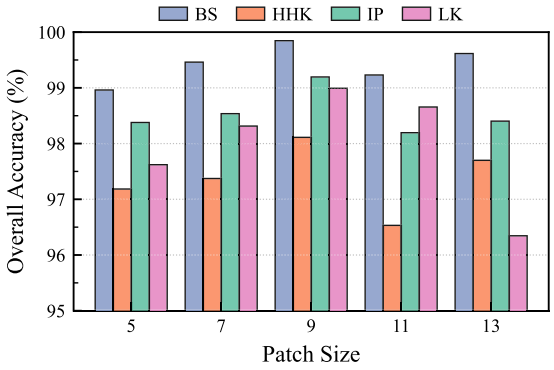


Fig. 7. Effect of different patch sizes on OA.

1) Patch Size: The patch size p defines the spatial neighborhood of the input centered at the labeled pixel, with a size of $p \times p$. When p is small, the available spatial contextual information may be insufficient. In contrast, when p is large, noise introduced by mixed pixels and boundary zero-padding is more likely to be incorporated. In addition, the number of tokens, i.e., $N = p^2$, increases with p, which results in a quadratic increase in the computational complexity of the attention mechanism. With all other hyperparameters fixed, p was evaluated at 5, 7, 9, 11, and 13, and the corresponding results are presented in Fig. 7. As p increases, the OA values on all four datasets first improve and then decline, with the best or near-best performance consistently observed at $p = 9$. Accordingly, $p = 9$ is adopted in all subsequent experiments.

2) Cutoff Ratio: The cutoff ratio τ controls the proportion of the low-frequency region suppressed during high-pass filtering in the 3-D-DCT domain. Since the DCT is performed along the spatial height, spatial width, and spectral dimensions, τ defines a normalized cutoff boundary in the 3-D DCT coefficient grid rather than a dataset-specific absolute threshold. Low-frequency components mainly describe smooth homogeneous regions, whereas mid-to-high-frequency components are more closely related to land-cover transitions, boundaries, textures, and mixed pixels. If τ is too small, excessively smooth responses are retained, which weakens local discriminative enhancement. If τ is too large, useful structural information may be discarded, and high-frequency noise may be amplified. With all other hyperparameters fixed, τ was evaluated from 0.2 to 0.8, and the results are shown in Fig. 8. For BS, IP, and LK, $\tau = 0.5$ achieves the best or near-best OA, while remaining competitive on HHK. Since τ defines a normalized cutoff boundary in the 3-D DCT domain rather than a dataset-specific absolute threshold, $\tau = 0.5$ provides a robust balance between suppressing redundant low-frequency responses and preserving discriminative mid-to-high-frequency information across diverse HSI scenes.

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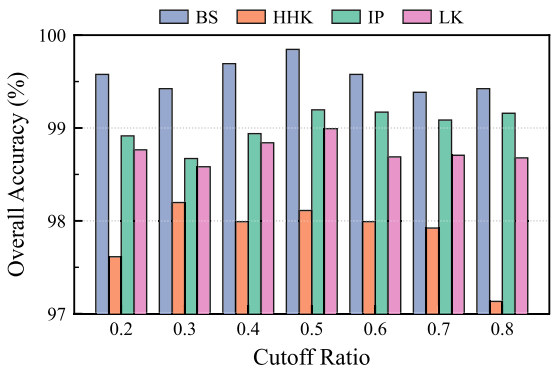


Fig. 8. Effect of different cutoff ratios on OA.

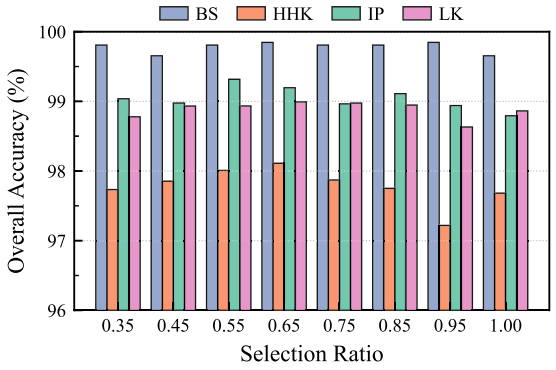


Fig. 9. Effect of different selection ratios on OA.

3) Selection Ratio: The selection ratio ρ determines the proportion of positions retained by the Selective Softmax mechanism. Specifically, only the top-k positions in the attention distribution of each query are preserved for normalization, where $k = \lfloor \rho N \rfloor$ and N denotes the number of spatial tokens (under the current experimental setting, $N = p^2$). When ρ is small, the attention becomes overly sparse and may fail to preserve critical contextual information. In contrast, when ρ is large, the attention mechanism approaches dense attention and is more likely to introduce redundancy and noise. With all other hyperparameters fixed, ρ was evaluated at 0.35, 0.45, 0.55, 0.65, 0.75, 0.85, 0.95, and 1.0, and the results are shown in Fig. 9. HHK and LK achieve their peak performance at $\rho = 0.65$, corresponding to $k = 52$ under the adopted 9×9 patch setting, after which the performance declines as ρ increases further. IP attains the highest OA at $\rho = 0.55$, while comparable performance is still maintained at $\rho = 0.65$. BS is relatively insensitive to variations in ρ. Considering the overall performance and stability across all datasets, $\rho = 0.65$ was uniformly adopted in the subsequent experiments. Overall, $\rho = 0.65$ provides a favorable balance between retaining informative contextual interactions and suppressing redundant correlations. Since the complete attention-logit matrix is computed before top-k masking, Selective Softmax does not change the asymptotic $\mathcal{O}(N^2)$ complexity in the current implementation; instead, it reduces the number of effective nonzero interactions and may offer further efficiency benefits with sparse-attention implementations.

D. Classification Results and Analysis

To evaluate the effectiveness of the proposed method, comparisons were conducted with nine representative baseline models. These baselines cover three mainstream categories of architectures, including purely CNN-based models (3-D-CNN and S2VNet), transformer-based models (SQSFormer, SiT, and CPFormer), and hybrid CNN-transformer architectures (SSFTT, HybridFormer, S2CIFTNet, and MCMTN).

1) Quantitative Result Analysis: Tables V–VIII summarize the performance of the proposed method and the competing methods on four datasets, namely, IP, BS, LK, and HHK. The reported metrics include OA, AA, κ, and class-specific accuracies. In each table, the best-performing results are highlighted in bold.

As shown in Table V, on the IP dataset, the proposed method achieves the highest OA of 99.195%, exceeding that of the second-best method, HybridFormer, by approximately 0.402 percentage points. Notably, for spectrally confusing and spatially mixed categories, such as Soybean-clean (593 samples) and Buildings-Grass-Trees-Drives (386 samples), the proposed method attains the highest class-specific accuracies of 97.263% and 98.382%, respectively. In comparison, SQSFormer achieves the second-best accuracy of 96.421% for Soybean-clean, while S2VNet and S2CIFTNet jointly achieve the second-best accuracy of 97.411% for Buildings-Grass-Trees-Drives. These results suggest that the proposed method exhibits stronger discriminative capability for complex spectral–spatial distributions.

Table VI reports the results obtained on the BS dataset, which is characterized by complex land-cover distributions. While 3-D-CNN yields an OA of only 77.260%, the proposed method achieves the highest OA of 99.846%, surpassing the best competing method, HybridFormer (99.307%), by 0.539 percentage points. Among the 14 classes, 11 reach 100% classification accuracy. Even for spectrally similar and easily confusable classes, such as Riparian, the proposed method maintains a high accuracy of 99.070%, demonstrating its effectiveness in distinguishing subtle spectral variations among complex land-cover categories.

As shown in Table VII, on the LK dataset, the proposed method achieves the highest OA of 98.992% with only 0.1% training samples, outperforming the second-best method, HybridFormer, by 0.475 percentage points. For the Sesame class, which contains only three training samples, FKCPA-Net still obtains a class-specific accuracy of 99.306%, indicating its effectiveness under extremely limited supervision. Nevertheless, the "Mixed weed" class shows a relatively lower accuracy of 88.578%. Although the frequency-aware enhancement strengthens boundary and texture representations, this class remains difficult due to its heterogeneous vegetation composition, ambiguous boundaries, and spectral similarity to neighboring crop and vegetation classes. This suggests that highly mixed and ambiguous vegetation categories remain challenging under scarce labeled samples.

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TABLE V
QUANTITATIVE PERFORMANCE OF DIFFERENT METHODS ON THE IP DATASET

Class No.	3D-CNN	SSFTT	HybridFormer	SQSFormer	S2VNet	MCMTN	SiT	S2CIFTNet	CPFormer	**FKCPA-Net**
1	**100.00**	86.486	**100.00**	**100.00**	97.297	**100.00**	97.297	97.297	**100.00**	**100.00**
2	67.192	89.151	**99.300**	97.900	95.801	97.900	96.238	98.250	97.638	99.038
3	72.590	87.801	99.548	98.795	99.398	97.289	98.645	99.699	99.548	**100.00**
4	64.737	92.632	98.421	95.263	97.368	96.316	88.421	**100.00**	99.474	97.895
5	89.896	94.041	97.409	95.855	96.891	98.187	97.668	97.409	**98.964**	98.446
6	96.918	**99.486**	97.603	98.288	**99.486**	98.801	97.432	**99.486**	98.801	**99.486**
7	68.182	90.909	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
8	96.597	**100.00**	99.215	99.738	**100.00**	99.215	98.168	**100.00**	**100.00**	**100.00**
9	93.750	56.250	93.750	**100.00**	**100.00**	93.750	81.250	93.750	**100.00**	**100.00**
10	72.751	91.517	98.715	97.943	98.329	96.915	97.686	**99.486**	95.887	98.458
11	81.008	96.741	**99.745**	98.880	98.320	99.287	97.301	98.727	99.033	99.542
12	72.421	85.053	94.947	96.421	96.000	94.105	92.211	92.842	94.316	**97.263**
13	98.780	98.171	**100.00**	99.390	**100.00**	99.390	99.390	**100.00**	99.390	**100.00**
14	96.838	99.111	99.209	99.605	99.605	99.308	99.704	99.901	**100.00**	99.802
15	73.463	92.233	96.764	95.146	97.411	92.880	96.764	97.411	96.117	**98.382**
16	29.730	58.108	**100.00**	98.649	**100.00**	**100.00**	**100.00**	93.243	**100.00**	98.649
OA (%)	80.659	93.573	98.793	98.244	98.171	98.012	97.183	98.610	98.378	**99.195**
AA (%)	79.678	88.606	98.414	98.242	98.494	97.709	96.136	97.969	98.698	**99.185**
$\kappa \times 100$	77.882	92.655	98.623	97.998	97.915	97.732	96.791	98.415	98.151	**99.082**

TABLE VI
QUANTITATIVE PERFORMANCE OF DIFFERENT METHODS ON THE BS DATASET

Class No.	3D-CNN	SSFTT	HybridFormer	SQSFormer	S2VNet	MCMTN	SiT	S2CIFTNet	CPFormer	**FKCPA-Net**
1	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
2	40.741	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	98.765
3	86.567	**100.00**	**100.00**	**100.00**	**100.00**	98.010	99.502	99.502	**100.00**	**100.00**
4	98.837	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
5	81.395	89.767	97.674	91.163	95.349	96.744	97.674	96.279	95.349	**99.535**
6	59.070	98.605	97.209	97.674	95.814	93.953	97.674	**100.00**	98.605	**99.070**
7	78.744	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
8	85.802	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
9	68.526	99.602	97.211	99.203	99.602	98.008	98.008	90.040	97.211	**100.00**
10	68.844	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
11	82.377	**100.00**	**100.00**	**100.00**	**100.00**	99.180	98.770	**100.00**	**100.00**	**100.00**
12	85.517	98.621	**100.00**	97.931	**100.00**	99.310	96.552	98.621	**100.00**	**100.00**
13	65.581	**100.00**	**100.00**	99.535	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
14	47.368	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	98.684	**100.00**	**100.00**	**100.00**
OA (%)	77.260	98.923	99.307	98.846	99.230	98.769	99.038	98.615	99.230	**99.846**
AA (%)	74.955	99.042	99.435	98.965	99.340	98.943	99.062	98.889	99.369	**99.812**
$\kappa \times 100$	75.327	98.833	99.250	98.750	99.166	98.666	98.958	98.500	99.166	**99.833**

TABLE VII
QUANTITATIVE PERFORMANCE OF DIFFERENT METHODS ON THE LK DATASET

Class No.	3D-CNN	SSFTT	HybridFormer	SQSFormer	S2VNet	MCMTN	SiT	S2CIFTNet	CPFormer	**FKCPA-Net**
1	81.343	97.939	99.697	99.364	99.152	99.509	98.705	95.741	99.080	**99.727**
2	44.394	96.159	94.448	44.370	81.297	90.858	96.111	**99.485**	93.766	98.432
3	34.479	61.620	92.397	92.165	93.025	96.860	96.463	96.529	90.380	**99.306**
4	83.748	91.989	97.931	92.635	97.234	97.253	97.608	98.897	98.684	**99.035**
5	31.958	80.014	**95.824**	87.521	75.091	94.931	87.280	69.491	87.280	95.728
6	39.704	98.073	99.704	96.086	99.341	99.341	98.199	99.560	99.087	**99.772**
7	99.148	99.933	**99.988**	**99.988**	99.867	99.483	99.607	99.883	99.736	99.656
8	77.932	87.848	95.288	91.927	97.243	96.512	93.418	97.103	97.032	**97.581**
9	77.520	90.993	**92.852**	89.134	86.355	80.682	83.154	79.686	80.050	88.578
OA (%)	82.084	95.258	98.517	94.180	97.101	97.722	97.679	97.565	98.029	**98.992**
AA (%)	63.358	89.396	96.459	88.132	92.067	95.048	94.505	92.931	93.899	**97.535**
$\kappa \times 100$	75.472	93.808	98.075	92.403	96.198	97.013	96.986	96.799	97.409	**98.677**

As shown in Table VIII, on the HHK dataset, the proposed method achieves the highest OA of 98.111%, outperforming the second-best method S2CIFTNet by 0.371 percentage points. For the challenging long-tail category, i.e., Class 20 (Tamarix, with 72 total samples and four training samples), the proposed method attains the highest class-specific accuracy of 93.846%, which is higher than those of S2CIFTNet (93.103%), CPFormer (91.379%), SiT (90.769%), and SQSFormer (87.692%). This result indicates that the proposed model maintains strong discriminative capability for long-tail distributions and spectrally confusing classes.

2) Visualization Results and Analysis: To qualitatively compare the classification performance of different methods, Figs. 10–13 present the classification maps for the

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TABLE VIII
QUANTITATIVE PERFORMANCE OF DIFFERENT METHODS ON THE HHK DATASET

Class No.	3D-CNN	SSFTT	HybridFormer	SQSFormer	S2VNet	MCMTN	SiT	S2CIFTNet	CPFormer	**FKCPA-Net**
1	85.028	**100.00**	92.655	97.458	98.588	96.328	92.655	92.038	97.452	**100.00**
2	**100.00**	99.721	**100.00**	99.860	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
3	78.788	88.889	66.667	93.939	88.889	74.747	94.949	86.364	98.864	**100.00**
4	75.439	92.982	93.567	90.643	**94.152**	91.228	89.474	90.789	89.474	91.813
5	36.000	68.000	44.000	61.333	45.333	52.000	84.000	77.273	**100.00**	93.333
6	46.512	82.558	95.349	91.860	89.535	98.837	94.186	98.701	**100.00**	91.860
7	29.412	69.412	52.941	**100.00**	50.588	54.118	**100.00**	**100.00**	97.368	**100.00**
8	73.684	97.368	99.474	**100.00**	**100.00**	**100.00**	97.895	98.817	97.633	**100.00**
9	99.444	**100.00**	93.889	**100.00**	98.333	92.778	**100.00**	99.375	93.750	**100.00**
10	97.981	92.043	**99.881**	99.287	99.406	99.525	98.575	99.332	97.463	99.762
11	99.398	99.799	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	99.774	**100.00**
12	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
13	97.846	**100.00**	**100.00**	99.692	97.846	99.692	**100.00**	99.308	**100.00**	**100.00**
14	72.222	84.259	89.815	87.500	**100.00**	94.907	90.741	92.188	94.792	93.519
15	99.254	99.067	99.440	**100.00**	99.813	99.627	99.067	**100.00**	96.639	**100.00**
16	99.756	**100.00**	**100.00**	**100.00**	98.533	99.756	**100.00**	**100.00**	**100.00**	99.022
17	57.500	81.667	69.167	68.333	68.333	76.667	68.333	**94.340**	77.358	68.333
18	90.560	90.560	96.755	93.215	94.395	97.345	85.546	**99.007**	98.013	96.165
19	68.852	81.967	75.410	80.328	65.574	65.574	78.689	83.333	83.333	**85.246**
20	30.769	75.385	58.462	87.692	83.077	76.923	90.769	93.103	91.379	**93.846**
21	34.286	60.000	45.714	97.143	42.857	82.857	**100.00**	90.323	96.774	**100.00**
OA (%)	89.990	94.900	94.797	96.841	95.725	95.879	96.343	97.740	97.450	**98.111**
AA (%)	74.892	88.747	84.437	92.775	86.441	88.234	93.566	94.966	95.718	**95.852**
$\kappa \times 100$	88.658	94.472	94.350	96.813	95.357	95.527	96.031	97.548	97.235	**97.951**

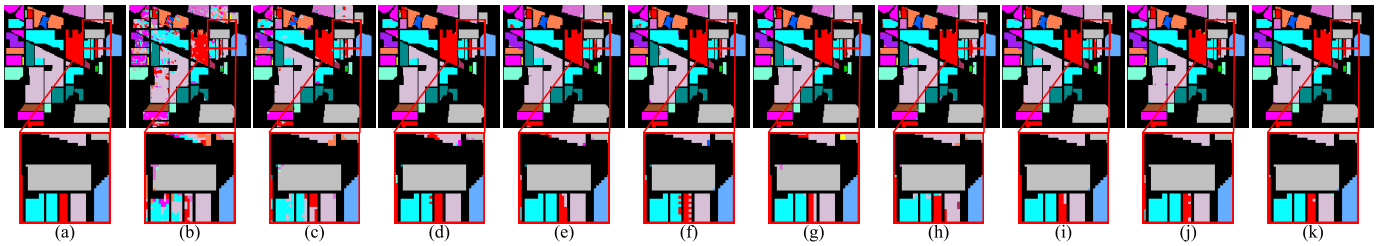


Fig. 10. Classification maps of different models on the IP dataset. (a) Ground truth. (b) 3-D-CNN. (c) SSFTT. (d) HybridFormer. (e) SQSFormer. (f) S2VNet. (g) MCMTN. (h) SiT. (i) S2CIFTNet. (j) CPFormer. (k) FKCPA-Net.

four datasets, namely, IP, BS, LK, and HHK. Specifically, (a) denotes the ground truth, (b)–(j) correspond to the nine competing methods, and (k) represents the proposed FKCPA-Net. As can be observed from the classification maps, FKCPA-Net produces results with lower noise levels and higher spatial continuity, and the boundaries of land-cover categories are more consistent with the ground-truth labels. In the magnified red-box region of the LK dataset, FKCPA-Net effectively suppresses salt-and-pepper noise within large homogeneous regions while accurately delineating the boundaries between adjacent classes. By contrast, methods such as 3-D-CNN exhibit evident salt-and-pepper noise in this region, and several transformer-based comparison methods suffer from boundary erosion or intraclass misclassification. Similar observations can also be made for the fine strip-like structures in the IP dataset, the complex mixed regions in the BS dataset, and the curved boundary regions in the HHK dataset. These results indicate that FKCPA-Net achieves a better balance between local-detail preservation and global-consistency modeling, thereby yielding classification maps with richer local details and spatial layouts that better match real land-cover patterns.

To explicitly locate misclassified pixels, Fig. 14 shows the error heatmaps of 3-D-CNN, HybridFormer, SiT, and FKCPA-Net on the four datasets, where red pixels indicate misclassified labeled pixels. Compared with 3-D-CNN, HybridFormer, and SiT, FKCPA-Net produces fewer scattered errors in homogeneous regions and fewer concentrated errors in the highlighted complex areas. Most residual errors are located near land-cover boundaries and mixed-pixel regions, indicating that FKCPA-Net can more effectively reduce misclassification errors while preserving spatial consistency.

3) Computational Cost Analysis: Table IX compares the parameters, FLOPs, model storage size, training time, and inference time of all methods on the four datasets. FKCPA-Net remains highly compact, with only about 22–23-K parameters and 0.11–0.12 MB storage across the four benchmarks. In terms of computational cost, FKCPA-Net achieves the lowest FLOPs on IP and BS, and comparable FLOPs on LK and HHK. The corresponding training times are 732.02, 230.71, 161.80, and 421.39 s on IP, BS, LK, and HHK, respectively. FKCPA-Net does not always achieve the shortest training or inference time, particularly on LK. Nevertheless, it combines high classification accuracy with a compact model and low computational complexity. Overall, the model provides a competitive balance among accuracy, storage requirements, and runtime efficiency.

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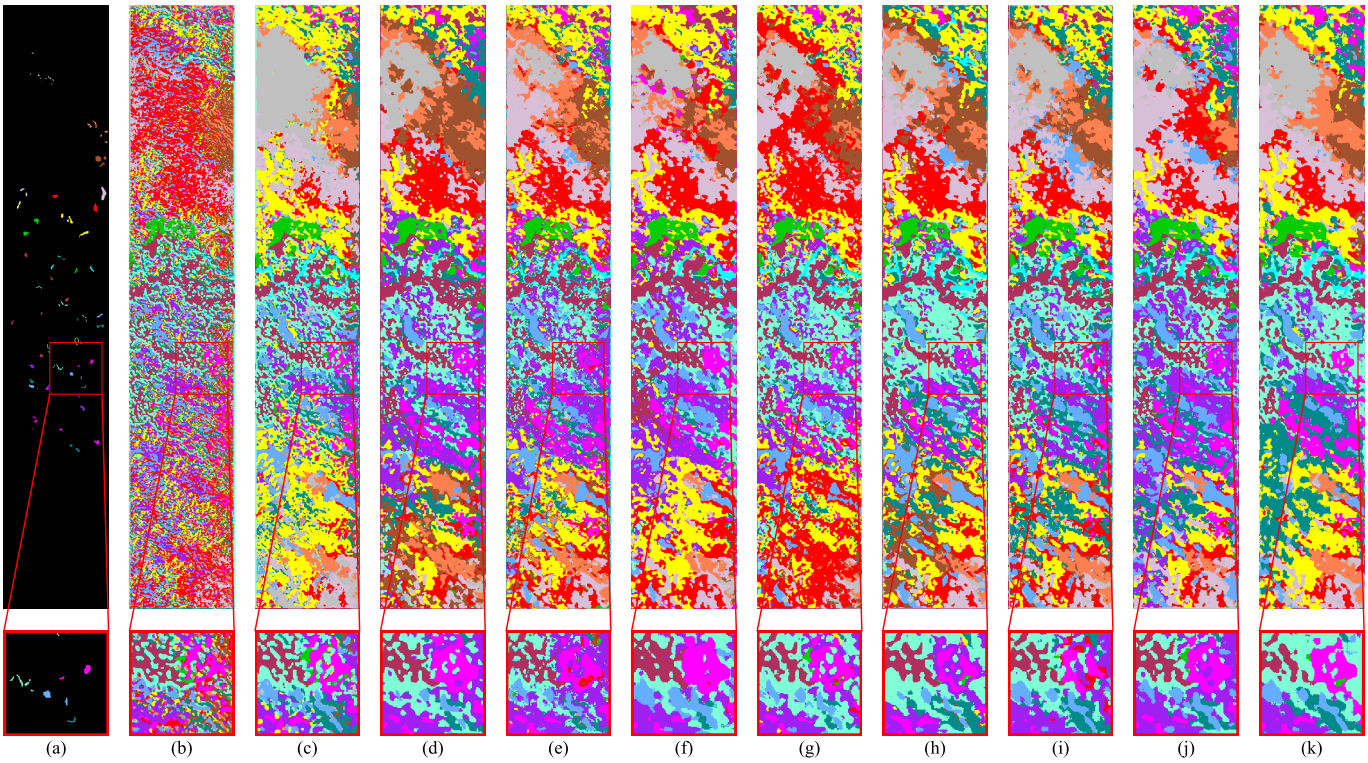


Fig. 11. Classification maps of different models on the BS dataset. (a) Ground truth. (b) 3-D-CNN. (c) SSFTT. (d) HybridFormer. (e) SQSFormer. (f) S2VNet. (g) MCMTN. (h) SiT. (i) S2CIFTNet. (j) CPFormer. (k) FKCPA-Net.

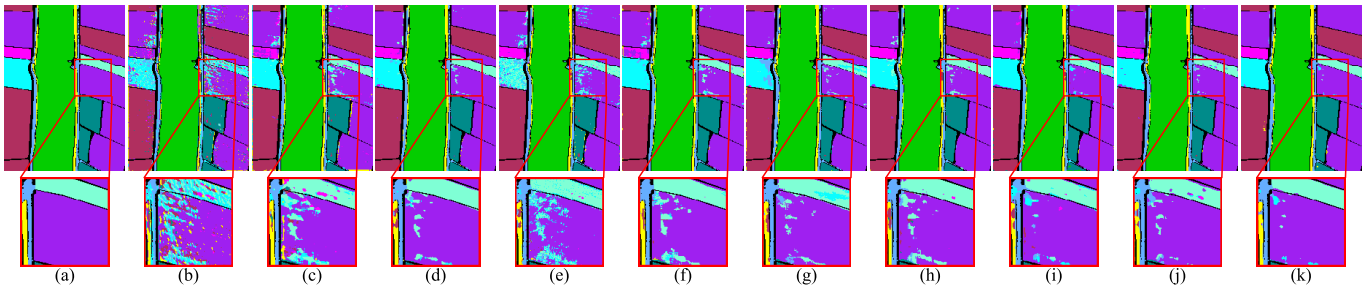


Fig. 12. Classification maps of different models on the LK dataset. (a) Ground truth. (b) 3-D-CNN. (c) SSFTT. (d) HybridFormer. (e) SQSFormer. (f) S2VNet. (g) MCMTN. (h) SiT. (i) S2CIFTNet. (j) CPFormer. (k) FKCPA-Net.

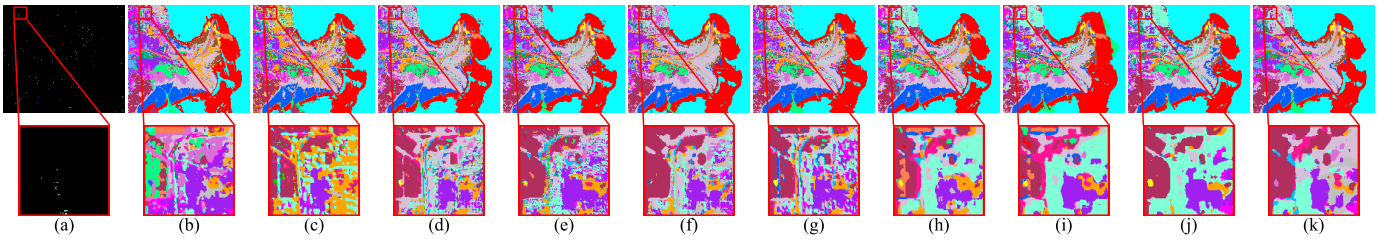


Fig. 13. Classification maps of different models on the HHK dataset. (a) Ground truth. (b) 3-D-CNN. (c) SSFTT. (d) HybridFormer. (e) SQSFormer. (f) S2VNet. (g) MCMTN. (h) SiT. (i) S2CIFTNet. (j) CPFormer. (k) FKCPA-Net.

4) Robustness Analysis: To assess robustness under different levels of supervision, multiple training-sample ratios were considered on all four datasets, and the corresponding OA, AA, and κ results are shown in Fig. 15. As the number of labeled data decreases, the performance of most baselines drops notably, whereas FKCPA-Net remains stable and competitive. Taking BS as an example, when the training ratio decreases from 10% to 2%, the OA of 3-D-CNN falls from 77.260% to 44.918%, whereas FKCPA-Net decreases only from 99.846% to 98.525% and remains above 98.00% throughout the tested range. On LK, FKCPA-Net still reaches 95.738% OA with only 0.05% training samples, and its OA

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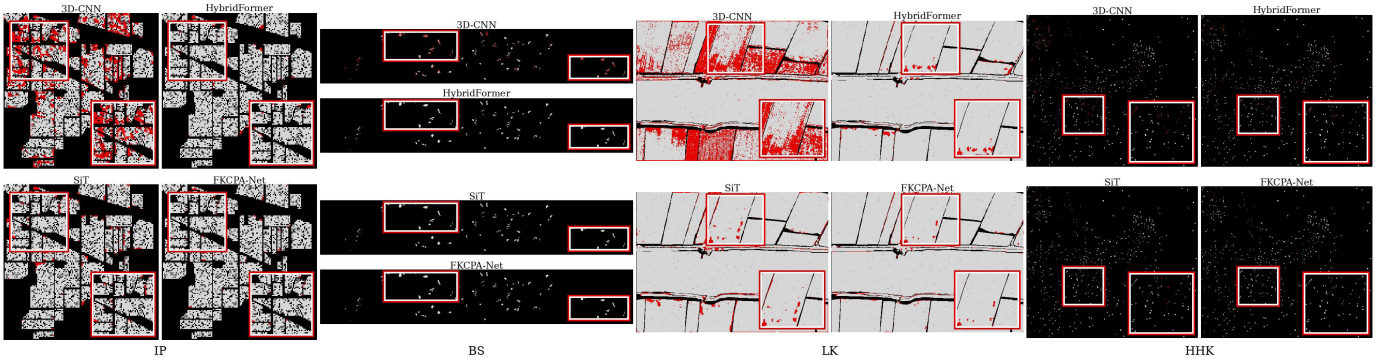


Fig. 14. Error heatmaps of 3-D-CNN, HybridFormer, SiT, and FKCPA-Net on the four datasets. Red pixels indicate misclassified labeled pixels.

TABLE IX
COMPARISON OF PARAMETERS, FLOPS, MODEL STORAGE SIZE, TRAINING TIME, AND INFERENCE TIME ON DIFFERENT DATASETS

Methods	IP					BS					LK					HHK				
	Params (K)	FLOPs (M)	Storage (MB)	Train (s)	Infer (s)	Params (K)	FLOPs (M)	Storage (MB)	Train (s)	Infer (s)	Params (K)	FLOPs (M)	Storage (MB)	Train (s)	Infer (s)	Params (K)	FLOPs (M)	Storage (MB)	Train (s)	Infer (s)
3D-CNN	394.64	68.98	1.51	**106.49**	0.97	263.38	50.05	1.00	155.18	0.22	298.38	92.95	1.14	123.55	32.26	692.14	98.85	2.64	234.91	0.98
SSFTT	931.85	25.31	3.57	173.60	0.27	683.01	18.08	2.61	44.98	0.06	1253.95	34.23	4.80	33.13	9.11	1328.59	35.69	5.07	63.46	0.27
HybridFormer	7210.83	124.05	27.70	439.14	1.00	6682.98	67.46	25.52	178.84	0.28	8197.75	204.81	31.50	113.80	34.40	8518.46	215.21	32.53	190.02	1.01
SQSFormer	258.52	20.10	1.05	134.28	0.21	243.70	17.54	0.93	**41.47**	0.07	297.94	23.37	1.20	25.06	5.39	325.70	24.07	1.24	**49.77**	0.15
S2VNet	321.01	33.72	1.23	153.49	**0.15**	209.84	20.75	0.80	51.17	**0.05**	456.49	40.82	1.74	32.26	4.96	549.58	65.15	2.10	58.85	0.15
MCMTN	221.10	13.84	0.85	134.62	0.19	205.00	12.56	0.79	50.55	0.06	240.36	15.48	0.93	**23.55**	**4.88**	246.22	15.83	0.93	55.59	**0.14**
SiT	567.54	42.44	2.18	290.92	1.13	474.58	34.94	1.82	144.52	1.13	717.15	54.82	2.75	60.71	30.49	759.53	57.89	2.92	125.73	0.89
S2CIFTNet	539.58	52.83	2.08	477.93	2.34	367.45	30.56	1.42	139.65	0.42	407.41	36.15	1.57	65.37	44.69	389.67	32.32	1.50	259.97	0.83
CPFormer	376.79	28.94	1.44	167.01	0.85	294.14	22.27	1.12	74.66	0.29	489.50	38.14	1.87	35.01	28.02	516.67	40.21	1.97	131.02	0.74
FKCPA-Net	**22.21**	**12.41**	**0.11**	732.02	2.42	**21.73**	**9.87**	**0.11**	230.71	0.57	**22.55**	15.89	**0.11**	161.80	80.08	**23.08**	16.67	**0.12**	421.39	2.51

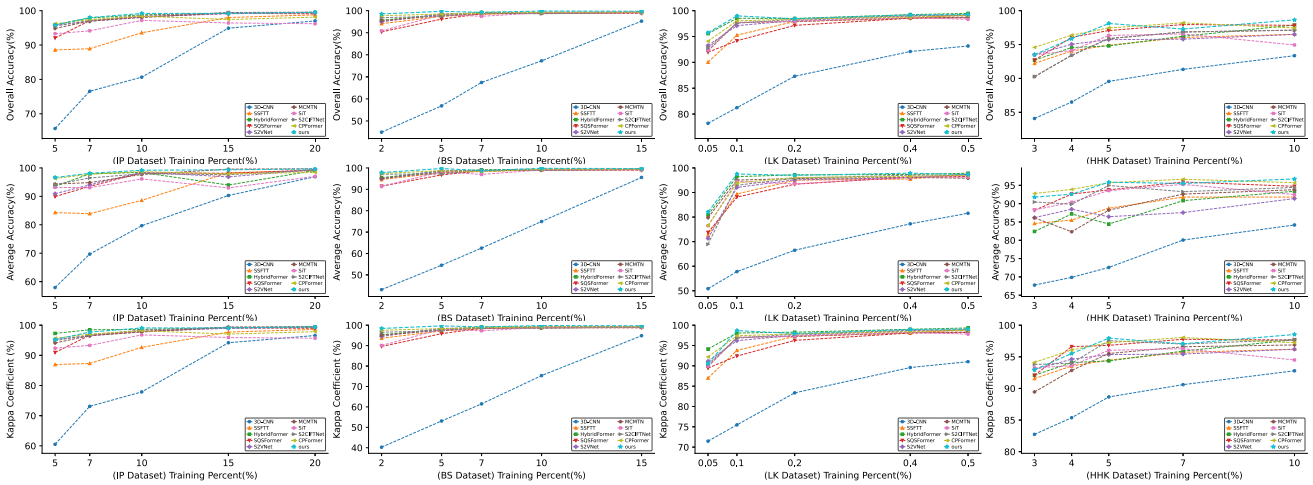


Fig. 15. Effect of different training sample proportions on OA, AA, and κ.

further increases to 98.992% when the training ratio is 0.1%. On IP, the proposed method maintains 95.816% OA with 5% training samples. On HHK, it still achieves 93.506% OA when only 3% of the data are used for training. In addition, the AA and κ results generally exhibit trends consistent with OA, indicating that FKCPA-Net also maintains competitive class-wise performance and reliable chance-corrected agreement under limited supervision. Although some competing methods obtain slightly higher values in isolated settings, FKCPA-Net consistently shows stable and competitive performance across different datasets, training ratios, and evaluation metrics. These results demonstrate that FKCPA-Net remains robust in low-label settings.

5) Ablation Study: To assess the contributions of the two key components, FASSF and KCPA, ablation experiments were conducted on all four datasets. Table X summarizes the results of different module combinations. Introducing either module individually improves classification performance over the baseline, while the full model consistently yields the best results. On the IP dataset, for example, the complete model reaches an OA of 99.195%, which is 0.402 and 0.280 percentage points higher than the variants using only FASSF or only KCPA, respectively, and 1.402 points higher than the baseline. Similar OA gains ranging from 1.077 to 1.425 are observed on the remaining datasets, and AA as well as κ follow the same trend. The gain differences observed across datasets are related to dataset-specific spectral–spatial characteristics. FASSF enhances local high-frequency cues such as boundaries, textures, and mixed pixels, while KCPA improves global contextual consistency through prior-guided attention. Their combination therefore yields complementary improvements.

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TABLE X

ABLATION ANALYSIS OF DIFFERENT COMPONENTS IN THE PROPOSED MODEL (BEST RESULTS IN BOLD)

Dataset	FASSF	KCPA	Evaluation Indicators (%)		
			OA	AA	κ
IP	×	×	97.793	97.660	97.482
	✓	×	98.793	99.025	98.624
	×	✓	98.915	99.080	98.630
	✓	✓	**99.195**	**99.185**	**99.082**
BS	×	×	98.769	98.742	98.666
	✓	×	98.961	99.183	98.875
	×	✓	99.077	99.079	98.999
	✓	✓	**99.846**	**99.812**	**99.833**
LK	×	×	97.828	93.384	97.145
	✓	×	98.153	95.699	97.575
	×	✓	98.220	96.867	97.669
	✓	✓	**98.992**	**97.535**	**98.677**
HHK	×	×	96.686	93.333	96.406
	✓	×	97.682	94.771	97.485
	×	✓	97.734	94.743	97.541
	✓	✓	**98.111**	**95.852**	**97.951**

TABLE XI

ABLATION STUDY OF KCPA SUBCOMPONENTS (BEST RESULTS IN BOLD)

Dataset	Kernel Centering	Distance Bias	Selective Softmax	Metrics (%)		
				OA	AA	κ
IP	✓	×	×	98.878	98.825	98.721
	×	✓	×	98.890	98.704	98.735
	×	×	✓	98.268	98.603	98.027
	✓	✓	×	98.756	98.779	98.582
	✓	×	✓	98.988	98.774	98.846
	×	✓	✓	98.976	99.005	98.832
	✓	✓	✓	**99.195**	**99.185**	**99.082**
BS	✓	×	×	99.423	99.458	99.375
	×	✓	×	98.961	99.103	98.875
	×	×	✓	99.115	99.231	99.041
	✓	✓	×	99.307	99.341	99.250
	✓	×	✓	99.538	99.441	99.500
	×	✓	✓	99.615	99.641	99.583
	✓	✓	✓	**99.846**	**99.812**	**99.833**
LK	✓	×	×	98.358	95.376	97.843
	×	✓	×	98.243	95.673	97.694
	×	×	✓	98.417	95.397	97.921
	✓	✓	×	98.484	96.597	98.011
	✓	×	✓	98.599	96.512	98.161
	×	✓	✓	98.645	96.807	98.222
	✓	✓	✓	**98.992**	**97.535**	**98.677**
HHK	✓	×	×	97.150	94.775	96.907
	×	✓	×	97.287	94.577	97.056
	×	×	✓	97.493	94.681	97.280
	✓	✓	×	97.665	94.494	97.467
	✓	×	✓	97.802	95.023	97.616
	×	✓	✓	97.579	93.955	97.373
	✓	✓	✓	**98.111**	**95.852**	**97.951**

In addition, we further conducted fine-grained ablation experiments inside the KCPA module to evaluate the individual effects of kernel centering, distance bias, and Selective Softmax, as reported in Table XI. It can be observed that removing any of these components causes performance degradation on all datasets. Specifically, kernel centering improves the modeling of relative token-correlation structures, the distance bias enhances spatially coherent attention allocation, and Selective Softmax suppresses weakly related token interactions during normalization. These results demonstrate that the three subcomponents are individually effective and complementary to each other.

The preceding ablation results show that FASSF and KCPA individually improve OA by 0.192–1.000 and 0.308–1.122 percentage points, respectively. Unlike DCT-based recalibration or center-pixel-guided attention, FASSF explicitly enhances 3-D spectral–spatial high-frequency cues, while KCPA regularizes the complete token-to-token correlation structure using structural and spatial priors. FKCPA-Net further outperforms CPFormer by 0.616–0.963 percentage points in OA across the four datasets, demonstrating their complementary benefits.

6) Attention Weight Visualization: To further verify the effects of the spatial-distance bias and Selective Softmax,

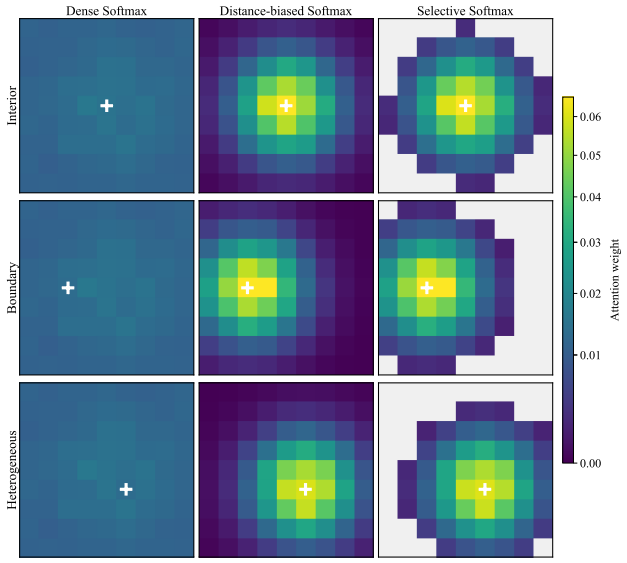


Fig. 16. Attention visualization of KCPA for interior, boundary, and heterogeneous regions. The columns compare dense softmax, distance-biased softmax, and selective softmax. The white cross denotes the query token, and white areas indicate masked tokens.

Fig. 16 visualizes the attention weights of representative query tokens selected from three typical regions, namely interior, boundary, and heterogeneous regions, in the KCPA module. Compared with Dense Softmax, the spatial-distance bias concentrates attention on spatially neighboring tokens around each query position, while Selective Softmax masks weak responses to zero. As a result, the retained attention weights form more compact and spatially coherent distributions. These results show that KCPA suppresses weakly related and spatially disconnected token interactions and promotes spatially coherent feature aggregation across different regional contexts.

V. CONCLUSION

This study proposes FKCPA-Net, a unified HSI classification framework that integrates multiscale spectral modeling, frequency-aware local enhancement, and prior-guided global attention. MSPeFE captures multiscale spectral dependencies, FASSF enhances boundary, texture, and mixed-pixel cues, and KCPA promotes spatially constrained long-range interactions through kernel-centered and distance-aware priors. Experiments on IP, BS, LK, and HHK demonstrate the effectiveness and robustness of FKCPA-Net under different scenes and limited supervision, while ablation studies verify the complementary contributions of FASSF and KCPA.

Nevertheless, extremely mixed or ambiguous categories remain challenging due to heterogeneous materials and strong spectral similarity to neighboring classes. Moreover, when applied to larger spatial patches or large-scale scenes, attention computation and frequency-domain operations may introduce additional computational overhead.

Future work will explore unsupervised domain adaptation for cross-scene and cross-sensor HSI classification. The FASSF module can be extended by aligning frequency-enhanced spectral–spatial features between source and target domains, while sensor-specific shallow projections can be used

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to map different spectral inputs into a common latent space before shared frequency-aware feature learning.

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